An urban signal-control task: model and sensitivity

1. Research questions

A signal can offer more left-turn service while a turner remains unable to reach the bay. The relevant obstruction depends on which traffic is in front, which lane it occupies, and whether another lane remains usable. The task asks how much of that ordered information is needed for useful signal-control sensitivities:

- **Information:** Can retained lane order explain reachable service beyond sound aggregate models using movement counts, storage and their own inference from traffic histories?
- **Decision:** Does that additional information change which effective green allocation reduces total time for the complete demand?
- **Gradient:** Does the predicted response to a small service reallocation agree with actual changes and help choose a better control?

These are separate claims: a more detailed state can improve a forward prediction without necessarily providing a more useful control direction.

2. Task and global assumptions

Control and population. Consider one isolated, initially symmetric four-leg junction with a finite set of releases and fixed routes. One constant vector $g \geq 0$, $\sum_{p=1}^{4} g_p = 1$, allocates effective service to NS through/right, NS left, EW through/right and EW left. Green fractions represent average service, rather than the active phase at a particular instant. Traffic is loaded from empty; any preparation period is the same for every candidate control. After the control window, a common fixed allocation clears all released demand.

Representative approach. There are two lanes before the bay opening and three at the stop line. Aggregate upstream road U feeds parallel local lanes F_1 (left/through) and F_2 (through/right). At the opening, F_1 branches into left-only bay B and through-only continuation T_1; F_2 continues into shared through/right lane T_2:

$$U \longrightarrow \{F_1, F_2\}, \qquad F_1 \longrightarrow \{B, T_1\}, \qquad F_2 \longrightarrow T_2.$$

The bay opening lies inside the local zone. Its cross-section also separates F_2 from T_2, so the two feeders cover the same upstream length. In the symmetric core B, T_1, T_2 have the same remaining length. Every section owns distinct road space and storage.

Left traffic enters F_1 and must use B; right traffic uses F_2, T_2. Through traffic can choose F_1 or F_2 when first admitted from U. Once in a feeder it stays in that lane: F_1 through traffic continues to T_1, and F_2 through traffic continues to T_2. The admission solver compares current feeder loads and feasible entry; it does not move resident traffic between lanes or predict a future signal plan. A blocked F_1 therefore need not stop traffic already using F_2. An occupied F_1 head still blocks traffic behind it. The smaller single-feeder layout remains a mechanism diagnostic; compared models always receive the same declared geometry and permissions.

Modeling assumptions.

- **MA1: continuous quantity, fixed routes.** Traffic mass is continuous vehicle-equivalent quantity. Routes, movements, lane permissions and topology are fixed metadata. Each realized lane retains an ordered record. Its cumulative-mass coordinate measures order, not physical metres.
- **MA2: average signal service.** Green fractions change movement service and downstream sharing. Road capacities, travel times, backward wave speeds, storage and saturation parameters are fixed. The model does not resolve red/green switching within a cycle.
- **MA3: physical access.** An unassigned input list determines which packet the allocator examines first. A packet already occupying a lane also has a physical position: traffic behind a blocked head cannot pass. Independent movement requires an independently reachable lane.
- **MA4: entry establishes lane occupancy.** Each packet is allocated using current receiving-lane loads and known physical limits. Later packets see earlier actual admissions. Unadmitted mass remains where it is, with no binding future-lane target and with priority over later packets of its movement.
- **MA5: causal timesteps.** Sending and receiving limits use histories available at the start of the update. New inflow cannot leave the same section in that update, and current departures do not instantly create entrance room. Boundary histories are interpolated linearly.
- **MA6: complete demand accounting.** Releases, routes and preparation are fixed across controls. Source waiting, moving traffic and queues all contribute until every released trip clears. Queue differences continue to matter after the control window ends.

Travel and storage. For each section e, capacity C_e is a mass rate and K_e is maximum stored mass. Free-flow travel time τ_e^f is the time to cross the section downstream. Backward-wave time τ_e^b is the time for a congestion or space-release effect to cross upstream. The triangular fundamental diagram relates these quantities, and the timestep must respect both travel directions:

$$K_e = C_e(\tau_e^f + \tau_e^b), \qquad 0 < \Delta t \le \min_e\{\tau_e^f, \tau_e^b\}. \tag{1}$$

For example, downstream departures release space, but the entrance can use that space only after the backward wave reaches it. Numerical lengths, demands and clearance horizons must satisfy this domain; they are not calibrated by the equations alone.

3. Traffic model

The model follows a traffic journey. First determine which resident mass has travelled far enough and where new mass can fit. Then assign accessible traffic to lanes. Stop-line heads use their movement's service and compete for downstream room. Finally update each physical section and retain the histories needed for the next timestep. Cumulative propagation and ordered route bookkeeping provide the backbone [1], [2]; the lane allocation and access rules below specify the local urban interaction.

3.1. Propagation and eligible arrivals. A section can send traffic only after it has travelled far enough, and can receive traffic only when room is available at its entrance. At $t_k = k\Delta t$, let $N_{e,k}^+$ count all earlier entries and $N_{e,k}^-$ all earlier departures. Their difference $n_{e,k} = N_{e,k}^+ - N_{e,k}^-$ is resident mass. The functions $\mathcal{H}_e^\pm$ read those cumulative histories between stored boundaries by linear interpolation: $\mathcal{H}_e^\pm((k+\theta)\Delta t) = (1-\theta)N_{e,k}^\pm + \theta N_{e,k+1}^\pm$. Histories are zero before empty loading.

The sending calculation first counts entries early enough to reach the exit by t_{k+1} under free flow, then subtracts mass that has already left. This gives eligible resident mass A. Only capacity $C_e\Delta t$ can cross during the update, so sending S is the smaller amount. Receiving uses the corresponding space account: start with storage K_e, add departures whose space-release effect can reach the entrance by t_{k+1}, and subtract all previous entries. The LTM limits are [1]

$$A_{e,k} = [\mathcal{H}_e^+(t_{k+1} - \tau_e^f) - N_{e,k}^-]_+, \qquad S_{e,k} = \min(C_e\Delta t, A_{e,k}), \tag{2}$$

$$R_{e,k} = \min\{C_e\Delta t,\, [K_e + \mathcal{H}_e^-(t_{k+1} - \tau_e^b) - N_{e,k}^+]_+\}. \tag{3}$$

Here $[z]_+ = \max(z, 0)$. In the first line, the delayed entrance history sets travel-time eligibility; in the second, the delayed exit history sets entrance-accessible room. Both S and R are mass allowances for this update, not instantaneous flow rates or guaranteed actual transfers. Signals, access and neighbouring sections can reduce realized flow. The timestep condition makes both delayed queries no later than t_k.

3.2. Lane allocation and physical access. The allocation problem is to admit as much of the current ready packet as feasible, then distribute that mass toward the less loaded legal lanes. Existing traffic is left in place. A packet is a homogeneous portion of continuous mass carrying a route reference, current movement and original entry/travel history [2].

For this packet, q is its eligible mass and s the sending owner's remaining transfer allowance. For each legal, reachable receiving lane, u_ℓ is its remaining admission allowance and h_ℓ its current load estimate: start-of-step resident mass plus earlier actual admissions. Thus V answers "how much can enter in total?" It is limited by the packet, the source and the combined receiving room. The lane solver then chooses a common filling level λ and additions x_ℓ:

$$V = \min(q, s, \sum_\ell u_\ell), \qquad x_\ell = \min\{u_\ell, [\lambda - h_\ell]_+\}, \quad \sum_\ell x_\ell = V. \tag{4}$$

The unknown λ is a queue-load level, measured in vehicle-equivalent mass. The gap $\lambda - h_\ell$ is how much that lane would need to reach the level. Add nothing to a lane already above it, and never exceed its individual bound u_ℓ. Choose the level so that all additions sum to V. This is the specified ideal lane-balancing rule, not a stop-line service cap. With initial loads $0.2, 0$, admitted mass $V = 0.6$ and nonbinding lane bounds, $\lambda = 0.4$ gives additions $0.2, 0.4$. A capped lane can remain below the common level. The additions are unique; λ itself can be nonunique when every usable lane is capped.

In the representative geometry, through packets have two alternatives at $U \to \{F_1, F_2\}$. Left and right packets have one legal feeder each, and movement fixes the branch at the bay opening. The same equation therefore covers both responsive choice and forced entry. Lane-entry room is separate from stop-line service: zero green does not immediately prevent a vehicle from joining a lane that it can reach and that has room.

Only admitted amounts establish new lane occupancy. Update the receiving loads by $h \leftarrow h + x$ and subtract those transfers from the remaining sending and receiving allowances. The remainder $q - V$ stays at its current location without a binding future-lane target. Inspect packets in recorded order; if one movement remains partly unserved, its later packets wait, while other movements with physical access can still be examined.

The sending allowance limits total mass actually admitted, not how many ready packets may be inspected. For example, an unassigned pool may list one blocked left-turn unit before a through unit with an independently reachable lane. With sending allowance one, inspecting only the first unit would miss the feasible through admission. Examining the through packet uses the same one-unit limit without moving it past a physical blocker. Unserved mass keeps its original entry/travel history and its relative position among the remaining entries; waiting does not restart its travel delay.

Physical lane order remains binding. A residual at the front of F_1 prevents following traffic from reaching either B or T_1, even if the allocator examines its record. Traffic already in F_2 has its own path to T_2. No traffic can jump from an occupied feeder into the parallel lane or enter B, T_1 directly from U.

3.3. Stop-line service and downstream sharing. Once a packet reaches the head of a stop-line lane, its movement sets how quickly it can discharge. Saturation rate $s_{\ell m}$ times green fraction $g_{p(m)}$ gives its average signal service, bounded by the physical lane capacity C_ℓ:

$$g_p \geq 0, \qquad \sum_{p=1}^{4} g_p = 1, \qquad \mu_{\ell m}(g) = \min\{C_\ell, s_{\ell m} g_{p(m)}\}. \tag{5}$$

For a current ready head set $c_\ell = \mu_{\ell m}$. Available service can still be reduced by the receiving road. Represent output j's timestep receiving allowance by the constant average rate $\rho_j = R_{j,k}/\Delta t$. If several lanes compete for that output, share first between inbound legs in proportion to their relevant green fractions, then equally between ready lanes of the same leg. A lane or leg unable to use its share releases that room to the other competitors.

Let $\mathcal{L}_{ij}$ contain current ready heads with positive remaining sending, from leg i to output j. Their combined cap is C_{ij}, their leg weight is w_{ij}, and W_j is the total rate that can be used. Common levels θ_j and η_{ij} are chosen to satisfy the respective rate totals:

$$C_{ij} = \sum_{\ell \in \mathcal{L}_{ij}} c_\ell, \quad w_{ij} = g_{p(i,j)}, \quad W_j = \min(\rho_j, \sum_i C_{ij}),$$

$$r_{ij} = \min(C_{ij}, \theta_j w_{ij}), \quad \sum_i r_{ij} = W_j, \tag{6}$$

$$r_\ell = \min(c_\ell, \eta_{ij}), \quad \sum_{\ell \in \mathcal{L}_{ij}} r_\ell = r_{ij}. \tag{7}$$

All quantities in these sharing equations are rates. The first level scales the green weights; the second fills uncapped lane rates equally. This uses the same capped-filling idea as lane allocation, but allocates *discharge rate* rather than new lane mass. Zero-cap heads do not compete, and zero receiving gives zero rates. Counting the green weight once per leg prevents an artificial priority gain from subdividing lanes. Shares apply to the heads ready now; later heads join when exposed.

Order determines when another movement becomes the head. Hold current rates until a head clears, a sending allowance is exhausted, or the update ends. With elapsed time σ, remaining head mass q_ℓ and remaining sending S_ℓ^{rem}, that next interval is

$$\delta = \min\left\{\Delta t - \sigma, \min_{\ell: r_\ell > 0} \frac{\min(q_\ell, S_\ell^{\text{rem}})}{r_\ell}\right\}. \tag{8}$$

During this interval, lane ℓ transfers $r_\ell \delta$. Then expose new heads and recompute rates, processing simultaneous events together. A zero-rate head blocks its own lane. This is an internal calculation of the timestep total; it neither introduces an active signal phase clock nor replaces the linearly interpolated propagation histories.

3.4. Mass conservation and travel histories. Every internal transfer must leave one physical section and enter the next with the same mass and route identity. If $b_{e,k}$ enters and $y_{e,k}$ leaves section e, update both cumulative counts and occupancy:

$$N^+_{e,k+1} = N^+_{e,k} + b_{e,k}, \quad N^-_{e,k+1} = N^-_{e,k} + y_{e,k}, \quad n_{e,k+1} = n_{e,k} + b_{e,k} - y_{e,k}. \tag{9}$$

The source first adds released mass $D_{o,k}$ to its waiting mass $Q_{o,k}$; admission then moves it onto the road. Sink exit removes it from the road population. The bounds $b \leq R \leq K - n$ and $y \leq S \leq n$ keep occupancy between zero and storage. Unadmitted mass is counted only where it physically remains. New admissions are appended after resident service has been selected.

The next update also needs to know when admitted mass entered. Rather than reconstruct a microscopic trajectory, spread the total entrance $b > 0$ uniformly across the timestep in its actual admission order. A child packet of mass q' starting at entrance rank r is assigned this entry-time interval [2]:

$$\left[t_k + \Delta t\,\frac{r}{b},\ t_k + \Delta t\,\frac{r+q'}{b}\right], \qquad \text{eligibility shifted by } \tau_e^f. \tag{10}$$

Here r is cumulative admitted mass before the child. The fractions r/b and $(r+q')/b$ locate its front and tail within the update; adding free-flow time gives its earliest exit times. No entry segment exists when inflow is zero. Internal section crossings retain route position; the junction crossing advances it. Exact-compatible adjacent records may merge while preserving route, order and timing. A movement grid summarizes mass along that order without replacing the exact head record.

4. How the task tests information and decisions

The question is whether two states with the same aggregate quantities can respond differently because their traffic is ordered differently. For a local bay-opening example, take ready F_1 traffic $[T(0.4), L(0.6)]$, sending allowance 1, and receiving allowances $R_{T_1} = 0.2$, $R_B = 1$. Only 0.2 through mass enters: its residual blocks left access. Reverse the order, keeping all lane loads and movement totals unchanged, and 0.6 left mass enters B before 0.2 through mass enters T_1. Meanwhile an independently ready 0.3 in F_2 can enter T_2 in either case. Total entry across the opening is therefore 0.5 versus 1.1: the obstruction is selective, while the parallel lane remains usable. This is a local arithmetic contrast, not yet proof of different full-trip time or failure of aggregate inference.

The control objective counts everyone still travelling or waiting, not only those who finish early. At each update charge one timestep to all available source mass and all road mass. Summing through a common horizon H beyond complete clearance gives total system time [1]:

$$J_{\Delta t} = \Delta t \sum_{k=0}^{H-1} \left[\sum_o (Q_{o,k} + D_{o,k}) + \sum_e n_{e,k}\right]. \tag{11}$$

Release $D_{o,k}$ is available before that update's admission, so it is included in that update's source charge. Transfers between road sections do not remove anyone from the population. This is a timestep quadrature; a nonclearing trip cannot disappear from a completed-trip-only score.

The scenario families separate simpler explanations before testing their interaction:

Case	Traffic contrast
V0	Movement-exclusive paths: matched propagation and service.
V1	Shared lane: head-order effects without a finite bay.
V2	Finite bay: storage with order-sensitive obstruction suppressed.
V3	The same bay geometry and permissions: order and access interact.
V4	Held-out variation of the declared families and conditions.

V2/V3 use the representative parallel-lane geometry. The smaller single-feeder case can isolate the mechanism, but cannot establish how an alternative lane moderates it. V0 equivalence requires matched service and nonbinding or equivalent receiving. A different propagation law or an unsuccessful storage model must be understood before attributing a V3 advantage to retained order. The shared through/right lane alone does not guarantee an order effect: its movements share green, and become service-distinct only when their effective caps or receiving differ.

Compare B0 aggregate, B1 structured movement queues, B2 responsive lane/storage aggregates and B3 selective ordered state under the same population, geometry, action and time account. These roles describe models, not a universal hierarchy of information. A macro model may read the original fine time series through its own converter or estimator. Its retained variables, memory, observation schedule and calibration must be declared. Successful aggregate inference is a valid explanation; an estimator retaining the full ordered ledger uses mesoscopic state. Equal current counts do not imply equal information in all histories. Distinguish a one-step test conditioned on observations, prediction with later observations, and forecasting without them; state which data remain fixed under a proposed service change.

The working B1 source is the link queue model [3]; the working B2 direction combines lane-exit allocation [4] with a sourced finite-storage/access adaptation, for which [5] is a candidate. Each complete baseline must be checked for physical soundness and task fit. Citation alone does not establish a valid bay-entry composition.

First test the same prescribed feasible service perturbations in all models. Then distinguish the value of the traffic representation from the value of an optimization algorithm by matching optimization budgets. A richer reference simulator uses one fixed mapping from effective service to executable signals, matched releases/routes, fixed calibration and predeclared seeds. Its population time includes insertion delay and travel duration [6],

including all unfinished demand in the account. Correct internal derivatives and a falling internal objective are insufficient to establish useful traffic-control guidance.

5. Complete sensitivity

Start with the traffic question: if a little service is moved from one group to another, which heads discharge more, which queues remain, and who can reach the bay next? The start-of-step state is already fixed; service first changes discharge and the next state. Queue and history changes then affect later admission, spillback and total time.

Consider a feasible change $g + \epsilon d$, with $\sum_p d_p = 0$. A dot denotes its derivative at $\epsilon = 0$, holding road parameters, routes and permissions fixed. The expressions below apply while the same heads and constraints remain active. A change in the blocking head or binding capacity can create a kink requiring one-sided treatment.

Lane-allocation response. A previous service change can alter both lane loads and the room available for entry. Among lanes receiving an uncapped positive addition, the common filling level must adjust so the additions still total V. Let $\mathcal{A}$ be these free lanes and $\mathcal{U}$ the lanes at their upper admission bounds. Differentiating the mass balance gives

$$\lambda = \frac{V - \sum_{\mathcal{U}} u_\ell + \sum_{\mathcal{A}} h_\ell}{|\mathcal{A}|}, \qquad \dot\lambda = \frac{\dot V - \sum_{\mathcal{U}} \dot u_\ell + \sum_{\mathcal{A}} \dot h_\ell}{|\mathcal{A}|},$$
$$\dot x_\ell = \begin{cases} \dot\lambda - \dot h_\ell, & \ell \in \mathcal{A}, \\ \dot u_\ell, & \ell \in \mathcal{U}, \\ 0, & \text{inactive lower branch.} \end{cases} \tag{12}$$

An uncapped lane responds through the common level minus its own load change; a capped lane follows its available room. Differentiate V through its limiting argument, then carry the additions into receiving loads, remaining allowances and retained mass $\dot q^{\mathrm{rem}} = \dot q - \dot V$ before processing the next packet. With no free lanes, use the cap or stable zero branches directly. A zero current admission does not erase the response of waiting mass. At a fixed pre-state this admission law has no direct green input; its control response arrives through the preceding traffic evolution.

Service and shared receiving. Green directly changes a head's service cap: $\dot\mu_{\ell m} = s_{\ell m} d_{p(m)}$ when signal-limited, and zero when physical capacity limits it. It also changes the leg's sharing weight. Receiving changes through previous road states, with $\dot\rho_j = \dot R_{j,k}/\Delta t$. The shared rate balance must reflect all of these changes. For uncapped legs $\mathcal{F}$ and capped legs $\mathcal{C}$, its response is

$$\dot\theta = \frac{\dot W - \sum_{\mathcal{C}} \dot C_{ij} - \theta \sum_{\mathcal{F}} \dot w_{ij}}{\sum_{\mathcal{F}} w_{ij}}, \qquad \dot r_{ij} = \begin{cases} \dot C_{ij}, & i \in \mathcal{C}, \\ w_{ij}\dot\theta + \theta \dot w_{ij}, & i \in \mathcal{F}. \end{cases} \tag{13}$$

Within a leg use the corresponding equal-share response. If no competitor is uncapped, follow cap branches without dividing by an empty weight sum.

Faster service can also expose the next head sooner. For an interior interval $\delta = q/r$, its time response is $\dot\delta = \dot q/r - q\dot r/r^2$. Total discharge changes through both the rates and the durations for which they apply:

$$y = \sum_a r_a \delta_a, \qquad \dot y = \sum_a (\dot r_a \delta_a + r_a \dot\delta_a). \tag{14}$$

Index a runs over internal service intervals. The final interval changes by the negative sum of the earlier duration changes so the whole update still lasts Δt. Freezing head-clearance times would omit this physical route by which one movement's service affects another.

For illustration, V1's shared left/through diagnostic has a ready left block q_L before sufficient through traffic. With nonbinding receiving and physical caps, left service first clears that block; through service uses only the remaining time:

$$y_T = s_T g_T \left(\Delta t - \frac{q_L}{s_L g_L} \right), \qquad 0 < \frac{q_L}{s_L g_L} < \Delta t. \tag{15}$$

With unit timestep and saturations, $q_L = 0.2$ and $(g_L, g_T) = (0.4, 0.6)$ give through discharge 0.3. Moving a small fraction from through to left has through-discharge derivative 0.25 on this branch: earlier head clearance outweighs the reduced through rate. This is a V1 service illustration, not left-turn permission in the representative through-only T_1, nor a claim about complete-time benefit.

Arrival timing and the full state. Earlier discharge also changes boundary counts and the entry times assigned to subsequent traffic. Differentiate the delayed histories and the reconstruction, so that later readiness changes with both packet mass and arrival time. For a positive-inflow branch the front-time response is

$$\dot t_{\mathrm{front}} = \Delta t \left(\frac{\dot r}{b} - \frac{r\dot b}{b^2} \right), \tag{16}$$

The tail has the analogous response. If fraction f of a packet's entry/travel segment is ready, ready mass responds as $f\dot{q} + q\dot{f}$. Zero inflow creates no timing segment.

Collect road masses, histories, ordered records and sources in state X_k, and write $P_k = \dot{X}_k$. If u_k is the green allocation applied in update k, the chain rule propagates the effect of a service change:

$$X_{k+1} = F_k(X_k, u_k), \qquad P_{k+1} = F_{k,X} P_k + F_{k,u} \dot{u}_k. \tag{17}$$

The first term carries earlier state differences forward; the second adds the current control change. Preparation is common and has zero candidate response. During control $\dot{u}_k = d$; after switching to fixed clearance $\dot{u}_k = 0$, but P_k continues to evolve. Switching the knob back does not undo the queue differences it created.

Finally, all released mass is either still waiting/travelling or has exited. With cumulative releases $D_{\le k}$ through update k and sink exits Z_k before that update, the same objective and its response can be written

$$J_{\Delta t} = \Delta t \sum_{k=0}^{H-1} (D_{\le k} - Z_k), \qquad \dot{J}_{\Delta t} = -\Delta t \sum_{k=0}^{H-1} \dot{Z}_k. \tag{18}$$

The sign is causal: earlier additional exits reduce population time. Use a common horizon that clears the compared control neighbourhood. This complete sensitivity is distinct from proving its numerical accuracy or downstream usefulness. Implementation, numerical derivative checks, baseline qualification and richer-reference tests remain necessary evidence. The mathematical companion gives full boundary cases and the derivation; source/adaptation records identify the origin of each operator.

References

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[6] SUMO official documentation, traffic-light programs and TripInfo population-time fields.